

The Transformer Decoder

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Sequence-to-sequence (encoder-decoder) ([Note 1068](#)); Transformer encoder ([Note 1080](#)); Masked self-attention ([Note 1081](#)); Cross-attention ([Note 1082](#)).
- **Leads to:** Transformer inference (autoregressive decoding, KV cache, beam search) ([Note 1084](#)); The transformer end to end (capstone) ([Note 1085](#)); Decoder-only GPT ([Note 1087](#)).
- **Compare with:** LSTM (long short-term memory) ([Note 1063](#)).

1. Overview

Key point: The **decoder** writes the output sentence. Its input is the target sentence **shifted right**: a start token in front, so that each position holds the word before the one it must predict. After embedding and positional encoding, the input passes through 6 identical **decoder blocks**. Each block has three sub-layers: **masked self-attention** over the output words so far, **cross-attention** to the encoder's output, and a feed-forward network, each followed by add and norm. A final **linear layer** with one node per word of the vocabulary, and a **softmax**, turn every position's vector into a probability for every word. During training all positions are computed in one pass.

The [transformer encoder Note](#) built the left half of the transformer of “Attention Is All You Need” (Vaswani et al. 2017). This Note builds the right half: the decoder. Two of its parts have their own Notes: [masked self-attention](#) and [cross-attention](#). Here they are put together with the rest into the full decoder (Figure 1).

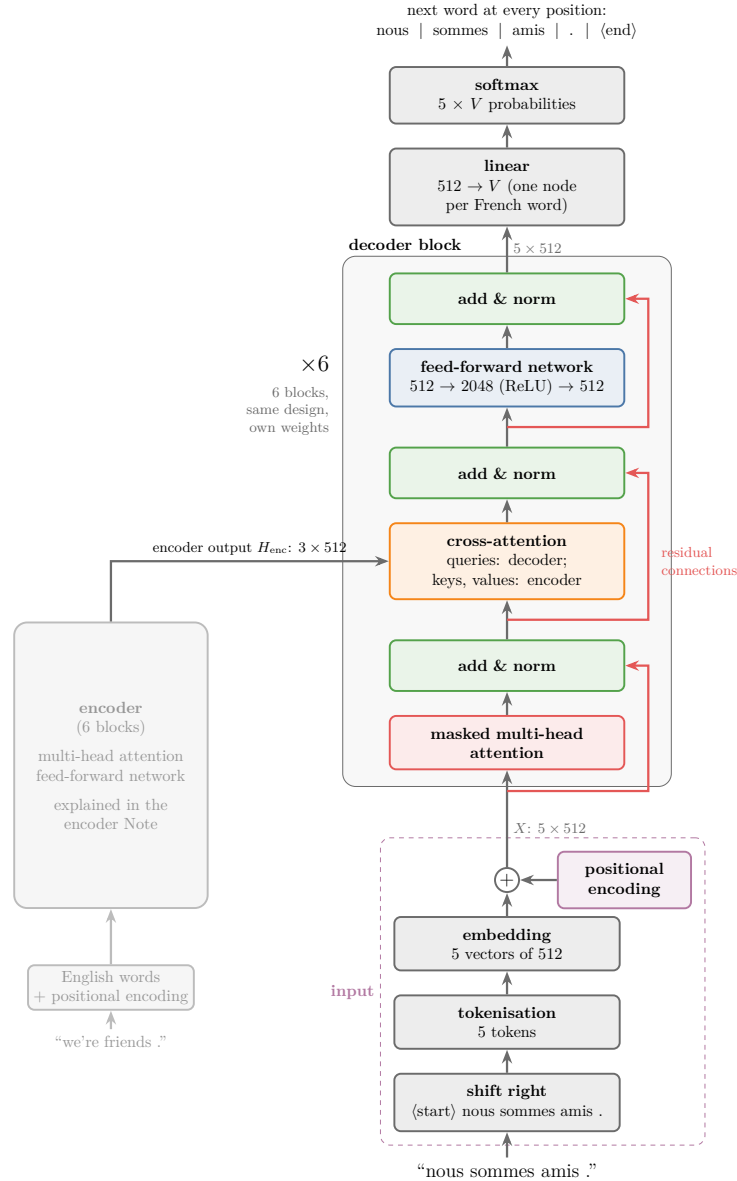


Figure 1: The transformer with the decoder in full. Bottom right: the input steps for the French sentence. Middle: one decoder block, repeated 6 times; red arrows are the residual connections. Left (grey): the encoder, whose output enters every cross-attention. Top: the output layer, which gives a probability for every French word at every position

The decoder behaves differently during training and during prediction. During training the whole correct output sentence is known, so all its positions are processed at once. During prediction the output is written one word at a time. This Note follows the **training** case; prediction is the subject of the [transformer inference Note](#).

We follow one real sentence pair from the English–French corpus of the Keras examples through the decoder: “we’re friends .” and its translation “nous sommes amis .”. Along the way we count shapes and parameters and check the counts against Keras.

2. Prerequisites

- [Transformer encoder Note](#): the encoder block, add and norm, the residual connection and the position-wise feed-forward network.
- [Masked self-attention Note](#): self-attention in which a word cannot attend to the words after it.
- [Cross-attention Note](#): attention with queries from one sequence and keys and values from another.
- [Encoder–decoder Note](#), section 5: <start> and <end> tokens, teacher forcing and the cross-entropy loss over the output steps.
- [Positional encoding Note](#) and [layer normalisation Note](#).

3. From a black box to the decoder block

Key point: The decoder is 6 identical blocks in a row. Each block has three sub-layers: masked multi-head self-attention, cross-attention, and a feed-forward network, with add and norm after each.

The decoder is built up in the same steps as the encoder.

1. **Two boxes.** A transformer is an encoder and a decoder. The encoder reads the input sentence; the decoder writes the output sentence.
2. **Six blocks.** “The decoder is also composed of a stack of $N = 6$ identical layers” (Vaswani et al. 2017, §3.1). We call each layer a **decoder block**. As in the encoder, the 6 blocks share a design and each has its own weights.
3. **Three sub-layers per block.** “In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack” (Vaswani et al. 2017, §3.1). In order, a decoder block holds:
 - **masked multi-head self-attention**: the output words attend to each other, each only to itself and the words before it;
 - **cross-attention** (also called encoder–decoder attention): the output words attend to the input words;
 - a **feed-forward network**, the same design as in the encoder.
4. **Add and norm.** Each sub-layer is wrapped in a residual connection followed by layer normalisation, as in the encoder.

Compared with an encoder block, the decoder block changes two things: its self-attention is masked, and it has one extra sub-layer, the cross-attention, which is the only place where the decoder reads the input sentence.

4. The setup: one training pair

Key point: Training data are sentence pairs. The encoder has already turned the English sentence into one vector per word, H_{enc} ; the decoder starts from the French sentence.

The data is a **parallel corpus**: each **observation** (one record of the data) is an English sentence and its French translation, the **target** (the output we want the model to produce). The Notebook uses the same filtered corpus as the [encoder–decoder Note](#): 40,000 training pairs, and a French vocabulary of the 8,000 most frequent words plus 4 special tokens, $V = 8,004$ in all. The pair we follow appears in the file as “We’re friends.” and “Nous sommes amis.”.

Before the decoder starts, the encoder processes “we’re friends .” exactly as in the [transformer encoder Note](#): 3 tokens, 6 encoder blocks, and out comes H_{enc} , a 3×512 matrix with one contextual vector per English word. The decoder now takes the French sentence.

5. The input: shift, tokenise, embed, add positions

Key point: Four steps prepare the French sentence. Shifting right puts `<start>` in front, so that position i holds the word before the one to be predicted at position i . Then tokenisation, a 512-number embedding per token, and the positional encoding, as in the encoder. The result is X , a 5×512 matrix.

5.1 Shifting right

Key point: The decoder’s input is the target sentence moved one place to the right, with `<start>` in the gap. Its target is the same sentence with `<end>` at the back.

The paper’s figure labels the decoder’s input “Outputs (shifted right)”. The **shift right** inserts the start token in front of the target sentence. Each position’s correct answer is then the next word:

Position	1	2	3	4	5
Decoder input	<code><start></code>	nous	sommes	amis	.
Target at that position	nous	sommes	amis	.	<code><end></code>

This offset is the transformer’s form of **teacher forcing**: during training the decoder is fed the correct previous words, not its own predictions (the [encoder–decoder Note](#), section 5.2, where the same shift appears). The paper states why the offset matters: the masking of the self-attention, “combined with fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i ” (Vaswani et al. 2017, §3.1). Without the shift, position 1 would see “nous” and simply copy it.

5.2 Tokenise, embed, add positions

Key point: Exactly the encoder’s three input steps, applied to the shifted French sentence.

1. **Tokenisation.** Each word and the full stop become a token: 5 tokens, with the ids 2, 21, 85, 250 and 4 in the French vocabulary.
2. **Embedding.** Each token becomes a learned vector of $d_{\text{model}} = 512$ numbers (the [word embeddings section](#) of the RNN sentiment Note). The French words have their own embedding table, $V \times 512$.
3. **Positional encoding.** The encoding of positions 1 to 5 is added to the embeddings (the [positional encoding Note](#)).

The result is the input matrix X , 5×512 : rows $x_1, \dots, x_5$.

Extra: The paper adds two details at this step (Vaswani et al. 2017, §3.4). The embedding vectors are multiplied by $\sqrt{d_{\text{model}}} = \sqrt{512} \approx 22.6$ before the positional encoding is added. And one weight matrix is shared by three layers: the encoder’s embedding, the decoder’s embedding and the final linear layer of section 8. The paper’s English–German data used one shared source–target vocabulary of about 37,000 tokens (§5.1). The Notebook applies the $\sqrt{d_{\text{model}}}$ factor; it keeps separate English and French vocabularies, so it does not share weights.

6. Inside one decoder block

Key point: $X \rightarrow$ masked attention $\rightarrow$ add & norm $\rightarrow Z_{\text{norm}}$; cross-attention with $H_{\text{enc}} \rightarrow$ add & norm $\rightarrow Z_{c,\text{norm}}$; feed-forward $\rightarrow$ add & norm $\rightarrow Y_{\text{norm}}$. Every one of these matrices is 5×512 .

Figure 2 shows the path of the five French positions through one block, with every shape.

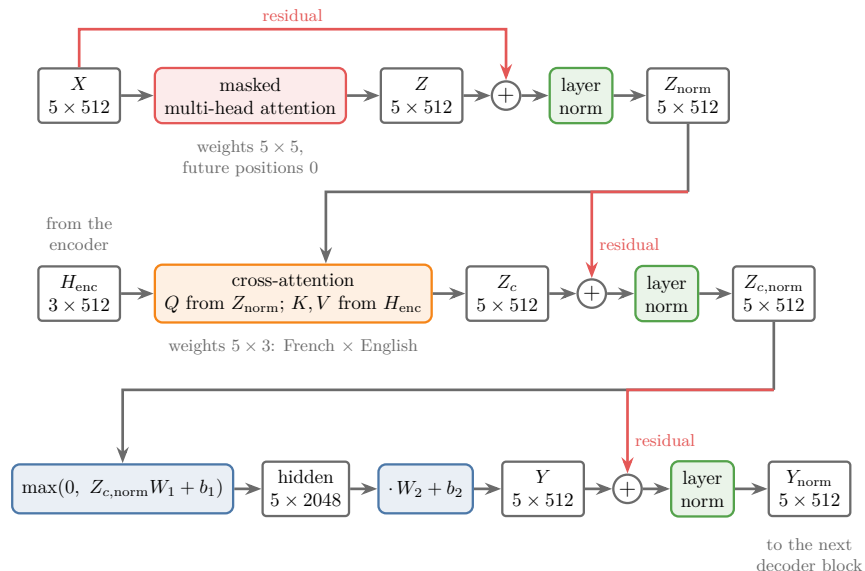


Figure 2: One decoder block for a 5-token French input and a 3-token English sentence. Every matrix is 5×512 except the encoder output H_{enc} and the hidden layer of the feed-forward network. Red arrows are the residual connections

Figure 3 follows a single position through a decoder block of a model that has been trained. It is the last decoder block of the small English-to-French transformer of the [transformer inference Note](#), with the decoder input `<start> nous sommes` and the English sentence “we’re friends .”. Watch what the position “sommes” may read at each sub-layer: earlier French words in the masked self-attention, the English words in the cross-attention (0.85 of its weight on “friends”), and nothing but itself in the feed-forward network. The output layer then predicts “amies” with probability 0.987.

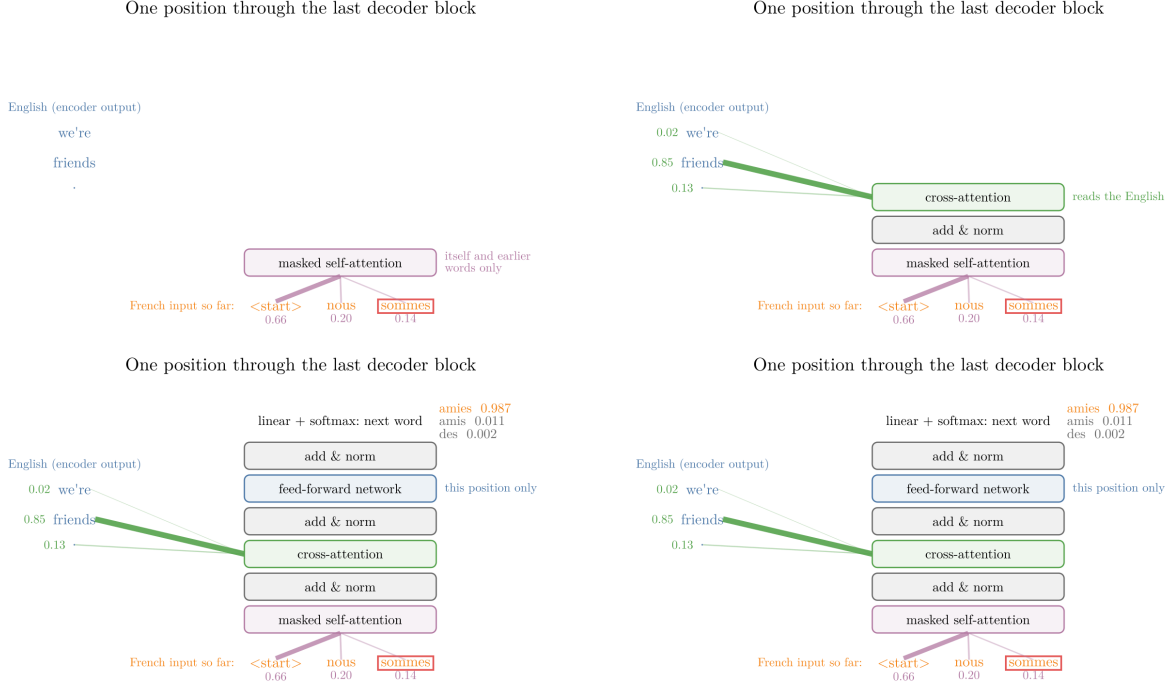


Figure 3: One position, “sommès”, through the last decoder block of a trained 2 + 2 block transformer. Line widths are the real attention weights (mean of 4 heads): masked self-attention over the French words so far, cross-attention over the English words, then the feed-forward network and the output layer

6.1 Masked multi-head self-attention

Key point: Multi-head attention among the French positions, with every weight on a later position set to exactly 0.

The first sub-layer is multi-head self-attention over X , with one change: a position may not look at the positions after it (the [masked self-attention Note](#)). For each row x_i it returns a contextual vector z_i :

- z_1 (for "<start>") uses only "<start>";
- z_2 (for "nous") uses "<start>" and "nous";
- z_3 uses the first three positions, and so on, up to z_5 , which uses all five.

The output Z is 5×512 , and the attention weights form a 5×5 table per head. In the Notebook’s untrained block, the weights averaged over the 8 heads are

Query \ key	<start>	nous	sommès	amis	.
<start>	1	0	0	0	0
nous	0.485	0.515	0	0	0
sommès	0.360	0.288	0.352	0	0
amis	0.258	0.239	0.259	0.243	0

Query \ key	<start>	nous	sommes	amis	.
.	0.225	0.170	0.168	0.158	0.280

Every entry above the diagonal is exactly 0, in all 8 heads. The sizes of the other weights mean nothing yet, since the block is untrained.

6.2 Add and norm

Key point: $Z_{\text{norm}} = \text{LayerNorm}(X + Z)$, as in the encoder.

The input X also takes the residual path around the attention and is added to its output, then each row is normalised: $Z' = X + Z$, $Z_{\text{norm}} = \text{LayerNorm}(Z')$. Both steps are exactly those of section 5.2 of the [transformer encoder Note](#). Z_{norm} is 5×512 .

6.3 Cross-attention

Key point: The queries come from the decoder (Z_{norm}); the keys and values come from the encoder’s output H_{enc} . Each French position gets a weighted sum of the English words’ value vectors.

The second sub-layer is where the two languages meet (the [cross-attention Note](#)). It is a multi-head attention with two inputs:

- **queries** from the decoder: $Q = Z_{\text{norm}}W_Q$, one per French position (5);
- **keys and values** from the encoder: $K = H_{\text{enc}}W_K$ and $V = H_{\text{enc}}W_V$, one per English word (3).

In the paper’s words, “the queries come from the previous decoder layer, and the memory keys and values come from the output of the encoder” (Vaswani et al. 2017, §3.2.3). The weights form a 5×3 table per head, French positions by English words, and no mask is needed: the whole English sentence is known. The output Z_c has one 512-number row per French position. In the untrained block of the Notebook every row of the averaged weights is close to $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$; training is what makes “amis” look at “friends”.

6.4 Add and norm

Key point: $Z_{c,\text{norm}} = \text{LayerNorm}(Z_{\text{norm}} + Z_c)$.

The input of the cross-attention, Z_{norm} , is added back to its output, and the sum is normalised. Note which matrix takes the residual path: the decoder’s own Z_{norm} , not the encoder’s H_{enc} . The shapes match, 5×512 plus 5×512 ; H_{enc} (3×512) could not be added.

6.5 Feed-forward network, add and norm

Key point: The encoder's position-wise network, $512 \rightarrow 2048$ (ReLU) $\rightarrow 512$, applied to each French position, then a third add and norm.

The third sub-layer is the same position-wise feed-forward network as in the encoder (the [transformer encoder Note](#), section 5.3), with its own weights. The 5 rows of $Z_{c,\text{norm}}$ go in together:

Step	Operation	Shape
hidden layer	$\max(0, Z_{c,\text{norm}}W_1 + b_1)$, W_1 : 512×2048	5×2048
output layer	$\text{hidden} \cdot W_2 + b_2$, W_2 : 2048×512	5×512
add and norm	$Y_{\text{norm}} =$ $\text{LayerNorm}(Z_{c,\text{norm}} + Y)$	5×512

Y_{norm} is the output of the block. The Notebook prints these shapes for the first of its 6 blocks: (5, 512) everywhere, (5, 2048) for the hidden layer, (8, 5, 5) for the masked weights and (8, 5, 3) for the cross-attention weights of the 8 heads.

Python: One decoder block as a Keras layer (post-norm, as in the paper). `use_causal_mask=True` applies the mask of section 6.1.

```
class DecoderBlock(keras.layers.Layer):
    def __init__(self, d_model, heads, d_ff):
        super().__init__()
        k = d_model // heads
        self.self_mha = keras.layers.MultiHeadAttention(num_heads=heads,
        ↪ key_dim=k)
        self.norm1 = keras.layers.LayerNormalization()
        self.cross_mha =
        ↪ keras.layers.MultiHeadAttention(num_heads=heads, key_dim=k)
        self.norm2 = keras.layers.LayerNormalization()
        self.ffn1 = keras.layers.Dense(d_ff, activation="relu")
        self.ffn2 = keras.layers.Dense(d_model)
        self.norm3 = keras.layers.LayerNormalization()

    def call(self, x, enc):
        z = self.self_mha(x, x, use_causal_mask=True)           # masked
        ↪ self-attention
        z_norm = self.norm1(x + z)
        zc = self.cross_mha(z_norm, enc)                       # queries:
        ↪ decoder; keys, values: encoder
        zc_norm = self.norm2(z_norm + zc)
```

```

        y = self.ffn2(self.ffn1(zc_norm)) #
    ↪ feed-forward network
    return self.norm3(zc_norm + y)

```

7. Six blocks: shapes and parameters

Key point: The shape stays 5×512 through all 6 blocks, and every block reads the same encoder output. One decoder block has 4,204,032 parameters, a third more than an encoder block because of the cross-attention; the 6 blocks have 25,224,192.

Shapes. The output of block 1 is the input X of block 2, which runs the same steps with its own weights, and so on up to block 6. Every block’s cross-attention receives the same H_{enc} , the output of the last encoder block (Vaswani et al. 2017, §3.1, “the output of the encoder stack”). The final output of the decoder is again 5×512 : one vector per French position. We call its rows $y_1, \dots, y_5$.

Parameters. With $d = 512$ and $d_{\text{ff}} = 2048$:

Part	Formula	Parameters
Masked multi-head attention	$4d^2 + 4d$ (the multi-head attention Note)	1,050,624
Cross-attention	$4d^2 + 4d$	1,050,624
Feed-forward network	$d \cdot d_{\text{ff}} + d_{\text{ff}} + d_{\text{ff}} \cdot d + d$	2,099,712
Three layer normalisations	$3 \times 2d$	3,072
One decoder block		4,204,032
6 decoder blocks		25,224,192

Keras’ `count_params()` gives exactly these numbers (Notebook). The cross-attention has the same parameter count as the self-attention: the encoder’s vectors have the same 512 numbers as the decoder’s, so W_K and W_V have the same shapes as before.

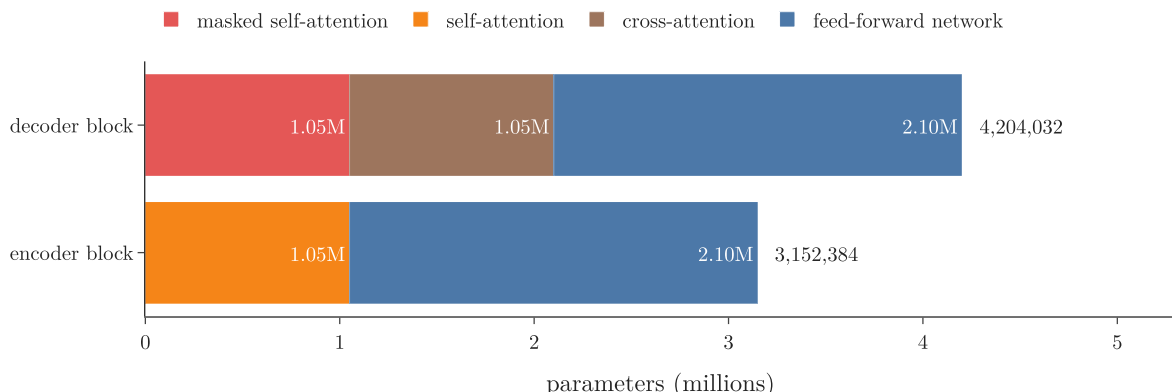


Figure 4: Parameters of one encoder block and one decoder block, by sub-layer. The layer norms (2,048 and 3,072) are too small to show

The whole base model. Adding the encoder’s 6 blocks (18,914,304, from the [transformer encoder Note](#)) and one shared $37,000 \times 512$ embedding matrix (section 5.2, Extra):

Part	Parameters
6 encoder blocks	18,914,304
6 decoder blocks	25,224,192
Shared embedding, about $37,000 \times 512$	18,944,000
Total	about 63 million

The paper lists 65 million parameters for its base model (Vaswani et al. 2017, Table 3), 2 million more than our 63 million. Table 3 itself shows where the gap sits. Its rows (C) change only the number of blocks N , so they separate the blocks from the rest:

N (Table 3)	Paper’s total	Blocks: $N \times 7,356,416$	Rest, outside the blocks	Rest $\div 512$
2	36 million	14.7 million	21.3 million	about 41,600
4	50 million	29.4 million	20.6 million	about 40,200
6 (base)	65 million	44.1 million	20.9 million	about 40,700
8	80 million	58.9 million	21.1 million	about 41,300

Here 7,356,416 is one encoder block plus one decoder block. Our block count agrees with the paper: from $N = 2$ to $N = 8$ the paper’s total grows by 44 million, 7.3 million per pair of blocks. The whole difference is in the rest, which is almost entirely the shared embedding: the paper’s totals fit a table of about 41,000 rows of 512 numbers (the totals are rounded to the nearest million, hence the spread), not the 37,000 we assumed from “a shared source-target vocabulary of about 37000 tokens” (§5.1). The paper does not give its exact vocabulary size, so we keep the 37,000 of §5.1 in the table above.

8. The output layer: linear and softmax

Key point: A dense layer with V nodes, one per word of the target vocabulary, turns each 512-number vector into V scores; the softmax turns the scores into probabilities that sum to 1.

After the sixth block, each French position has a 512-number vector, and we need a word. The output layer has two parts (the “Linear” and “Softmax” boxes at the top of Figure 1):

1. **Linear.** A dense layer with no activation, 512 inputs and V nodes, one node per word of the French vocabulary. Its weights W_3 are $512 \times V$ plus V biases. Its outputs, one unnormalised score per word, are called **logits**.
2. **Softmax.** The softmax turns the V logits of each position into V probabilities that sum to 1 (the [loss functions Note](#)).

The **vocabulary** is the list of all distinct words of the French side of the data. A larger vocabulary means more nodes: with the Notebook’s $V = 8,004$, the layer has $512 \times 8,004 + 8,004 = 4,106,052$ parameters, about as many as one decoder block.

1. **In words:** multiply each position’s vector by W_3 and add the bias to get one score per word, then exponentiate each score and divide by the sum of the exponentials.
2. **Formula:** for the vector y_i of position i ,

$$u_i = y_i W_3 + b_3 \quad (1 \times V), \quad P(\text{word } k \text{ at position } i) = \frac{e^{u_{i,k}}}{\sum_{j=1}^V e^{u_{i,j}}}$$

3. **Example:** a toy vocabulary of 4 words, nous, sommes, amis, <end>, with logits 2.0, 1.0, 0.5, -1.0 at position 1:

$$e^{2.0} = 7.39, \quad e^{1.0} = 2.72, \quad e^{0.5} = 1.65, \quad e^{-1.0} = 0.37, \quad \text{sum} = 12.13$$

$$P = \frac{7.39}{12.13}, \frac{2.72}{12.13}, \frac{1.65}{12.13}, \frac{0.37}{12.13} = 0.609, 0.224, 0.136, 0.030$$

“nous” has the highest probability, which is correct for position 1.

All 5 positions go through the same layer together: 5×512 times $512 \times V$ gives a $5 \times V$ table of probabilities, one row per position. In the Notebook every row sums to 1.

9. Training: every position in one pass

Key point: One forward pass produces a probability row for every position at once. The loss is the cross-entropy between each row and the target word of that position, averaged over the positions. No word is chosen during training.

Because the whole target sentence is known and shifted right, and the mask stops each position from seeing later ones, all 5 predictions can be made in the same pass (the [masked self-attention Note](#)). The decoder is **non-autoregressive** during training: it does not wait for its own outputs.

The five rows are then compared with the five targets of section 5.1. As in the encoder–decoder (section 5.3 of the [encoder–decoder Note](#)), each position is a classification over the vocabulary, and the loss is the categorical cross-entropy, minus the log of the probability given to the correct word (SLP3 eq. 13.13):

$$L = -\frac{1}{5} \sum_{i=1}^5 \ln P(t_i \text{ at position } i)$$

where t_i is the target word of position i .

Before training, the model has no preference: each correct word gets about $1/V$ of the probability. In the Notebook the 5 target words get $0.76/V$, $0.57/V$, $0.79/V$, $0.38/V$ and $1.04/V$, and the loss is 9.39, close to $\ln V = \ln 8,004 = 8.99$. Keras’ `SparseCategoricalCrossentropy` gives the same 9.39. Backpropagation then updates every weight of the decoder, the cross-attention’s W_K and W_V , and through them the whole encoder, as one network.

The untrained decoder’s most likely word is “mit” at all 5 positions. Picking the most likely word is not part of training: the loss uses the probability of the correct word, whatever word ranks first. Choosing words happens only during prediction, where the decoder works **autoregressively**, one word per step, each new word becoming part of the next input (the [transformer inference Note](#)).

10. Summary

Step	Operation	Shape (3 English, 5 French tokens)
Encoder	6 encoder blocks	$H_{\text{enc}}: 3 \times 512$
Input	shift right, tokens $\rightarrow$ embeddings + positional encodings	$X: 5 \times 512$
Sub-layer 1	masked multi-head self-attention, then add & norm	$Z_{\text{norm}}: 5 \times 512$ (weights 5×5)
Sub-layer 2	cross-attention to H_{enc} , then add & norm	$Z_{c,\text{norm}}: 5 \times 512$ (weights 5×3)
Sub-layer 3	feed-forward network, then add & norm	5×2048 inside, $Y_{\text{norm}}: 5 \times 512$
Repeat	6 blocks, own weights, same H_{enc}	5×512
Output	linear ($512 \rightarrow V$) and softmax	$5 \times V$ probabilities

- The decoder’s input is the target sentence shifted right: **<start>** in front, so position i predicts word i from the words before it.
- A decoder block is an encoder block with a masked self-attention and an extra cross-attention sub-layer; it has 4,204,032 parameters against 3,152,384.
- Cross-attention takes its queries from the decoder and its keys and values from the encoder’s final output, in every block.
- The output layer has one node per vocabulary word; the softmax gives a probability for every word at every position.
- During training all positions are computed in one pass and the loss is the average cross-entropy; during prediction the decoder writes one word at a time.

11. Sources

Built from

- CampusX, “Transformer Decoder Architecture | Deep Learning | CampusX”, YouTube, https://www.youtube.com/watch?v=DI2_hrAulYo

Other references

- Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS 2017*. arXiv:1706.03762. §3.1 (decoder: $N = 6$, the third sub-layer, the masking and the offset by one position); §3.2.3 (encoder–decoder attention: queries from the decoder, keys and values from the encoder); §3.4 (shared embedding and pre-softmax weights, the factor $\sqrt{d_{\text{model}}}$); §5.1 (shared vocabulary of about 37,000 tokens); Figure 1 (“Outputs (shifted right)”); Table 3 (base model, 65 million parameters; rows (C), totals for $N = 2, 4$ and 8).

- Jurafsky, D. and Martin, J. H. *Speech and Language Processing*, 3rd ed. draft (19 August 2026), ch. 13, §13.3 (the encoder–decoder transformer, cross-attention, eq. 13.11–13.13, teacher forcing).

12. Key terms

Term	Meaning
Decoder	The part of the transformer that writes the output sentence, using the encoder’s output
Decoder block	One layer of the decoder: masked self-attention, cross-attention and a feed-forward network, each followed by add and norm; the decoder stacks 6
Observation	One record of the data; here, one English sentence and its French translation
Target	The output we want the model to produce; here, the French sentence
Shift right	Putting <start> in front of the target sentence to form the decoder’s input, so each position predicts the next word
Teacher forcing	Feeding the decoder the correct previous words during training instead of its own predictions
Masked self-attention	Self-attention in which each position attends only to itself and earlier positions
Cross-attention	Attention with queries from the decoder and keys and values from the encoder’s output; also called encoder–decoder attention
H_{enc}	The output of the last encoder block: one vector per input word, read by every decoder block
Vocabulary (V)	The list of distinct words (tokens) of the target language; the output layer has one node per word
Logit	The unnormalised score the linear layer gives a word, before the softmax
Non-autoregressive	Producing all output positions at once, as the decoder does during training
Autoregressive	Producing one output at a time, each fed back as input for the next, as the decoder does during prediction